

Analytical modelling of membrane polarization with explicit parametrization of pore radii and the electrical double layer

Matthias Bucker^{1,2} and Andreas Hördt¹

¹*Institute of geophysics and extraterrestrial physics, TU Braunschweig, Mendelssohnstr. 3, 38106 Braunschweig, Germany.*

E-mail: a.hoerd@tu-braunschweig.de

²*Department of Geodynamics and Geophysics, University of Bonn, Bonn, Germany*

Accepted 2013 April 4. Received 2013 April 4; in original form 2012 November 27

SUMMARY

Processes related to the membrane-like behaviour of pore throats are generally considered relevant for the induced polarization of granular media. A model to describe the impedance of membrane polarization analytically was suggested previously for a 1-D system, consisting of a sequence of narrow and wide pores of different lengths. In that model, the membrane effect is caused by different mobilities of the ion species, resulting in different ion transference numbers in the two pore types. We extend the model by explicitly including the properties of the electrical double layer and the pore radii into the impedance calculation. Our basic assumption is that in a cylindrical pore system, the ion concentrations in the direction perpendicular and parallel to the pore are independent of each other. We then obtain concentrations averaged over the pore section from known theories of the charge distribution in a cylindrical pore. We also include the effect of the Stern layer in form of a partition coefficient. Our extension enables us to use the analytical expression of the impedance, but with an extended parametrization of the transference numbers. We demonstrate the validity of our approach by comparing the results of our calculations with previously published numerical simulations for a particular model. The usefulness of our method is illustrated with parameter studies that illustrate the possible dependence of relaxation times on pore radii.

Key words: Numerical solutions; Electrical properties; Hydrogeophysics; Permeability and porosity.

INTRODUCTION

The potential of the spectral induced polarization (SIP) method to estimate hydraulic conductivity of sediments has been widely recognized. However, there is no general agreement yet which parameters derived from SIP measurements should be used. Some authors suggest to base hydraulic conductivity estimation on a characteristic timescale derived from the spectral shape of the complex conductivity (Kemna *et al.* 2005; Revil & Florsch 2010; Weller *et al.* 2010a; Revil *et al.* 2012). The alternative is to use the imaginary conductivity obtained at a single-frequency to derive the specific inner surface area, from which hydraulic conductivity may be estimated (e.g. Weller *et al.* 2010b). This approach, because of its small required bandwidth, was already applied at the field scale (e.g. Kemna *et al.* 2004; Hördt *et al.* 2009). Both methods have their theoretical justifications and were demonstrated to have some predictive abilities with laboratory data, yet the underlying concept seems to be quite different.

The uncertainty which parameters are suitable and in which form is partly because the dominating processes at the pore scale are not yet fully understood. Two main processes are being discussed: The electrochemical polarization, which is based on charge motion

within the electrical double layer (EDL) at the surface of the mineral grains, and the so-called membrane polarization, caused by narrow throats within the pore space.

The theory of electrochemical polarization, first suggested by Schwarz (1962) to describe the complex conductivity of a spherical grain, was recently extended to derive equations for more complex grain size distributions and relaxation times (Leroy *et al.* 2008; Leroy & Revil 2009; Revil & Florsch 2010). The theory is well developed and allows a number of geometrical and electrochemical parameters, such as grain size distribution, pH, ion valence and temperature, to be included in the model. The theory of membrane polarization, suggested by Marshall & Madden (1959), is based on the model of a sequence of narrow and wide pores, in which a variation of ion mobilities causes an ion concentration gradient that leads to a frequency-dependent conductivity.

The two processes (electrochemical polarization and membrane polarization) do not exclude each other, and may both be relevant. Some authors have suggested to merge them in a single model (e.g. Leroy & Revil 2009; Jougnot *et al.* 2010), but the implementation is not trivial and no breakthrough was obtained yet. Lacking such a comprehensive model, there is an ongoing debate whether one of the processes dominates the spectra observed in real measurements

(e.g. Revil *et al.* 2012). This question seems to be difficult to answer, partly because the theories generate similar falsifiable hypotheses, such as the relationship between spatial scales and relaxation times (e.g. Blaschek & Hördt 2009). Few experiments have been designed explicitly to decide between the two theories, but even then the interpretation of the results is complicated and it is difficult to arrive at definitive conclusions (e.g. Scott 2006; Hördt & Milde 2012). However, it is of course important to identify the dominating processes and to advance the corresponding theories, because even minor differences in the equations may lead to significant differences in the prediction of hydraulic parameters (e.g. Koch *et al.* 2012; Revil *et al.* 2012).

The present paper deals with the membrane polarization theory, which may be considered less developed than the electrochemical polarization, in the sense that it does not yet include the properties of the EDL. It is based on the assumption that anion and cation mobilities are different in narrow pores, and the geometry is described in terms of the lengths of pores, and not their widths (e.g. Marshall & Madden 1959). As a result, discussions of membrane polarization are usually based on the pore length as the key parameter, and existing ideas to estimate hydraulic conductivity from pore radii distributions (e.g. Juang & Holtz 1986) could not yet be fully exploited. Sometimes, the assumption is made that pore widths have the same spatial scale as the pore length (e.g. Titov *et al.* 2010). On the contrary, for electrochemical polarization, a complete theory to determine hydraulic conductivity from SIP measurements for granular media has been suggested, which is based on the convolution of a single grain size response with the grain size probability distribution (Revil & Florsch 2010).

Some extensions of the original model by Marshall & Madden (1959) have been suggested to include the pore section and the EDL properties as additional parameters into membrane polarization theory. Buchheim & Irmer (1979) included the pore section explicitly into the boundary conditions. They also suggested to use averaged ion concentrations to calculate ion transference numbers, without, however, carrying out the corresponding derivations. Sidhorova & Fridrikhsberg (1978) consider the pore section in form of an efficiency parameter. Titov *et al.* (2002) also use ion concentrations averaged over the pore section to calculate ion transference numbers, without, however, accounting for the influence of the EDL on the ion distribution within the pore. A finite pore size was considered explicitly by Volkmann & Klitzsch (2010), who carried out numerical simulations to study the dependence of the SIP spectra on geometrical parameters and water salinity. We use their results below for comparison with our analytical solutions.

Here, we include the properties of the EDL by taking into account the ion concentration as function of distance from the pore surface when averaging over the radius. We explicitly consider the Stern layer (SL) in form of a partition coefficient as suggested by Leroy *et al.* (2008). Our approach allows us to use the analytical solution originally developed by Marshall & Madden (1959) for a 1-D pore system, with pore radii as additional parameters. This opens up a whole new field of possibilities to simulate membrane polarization and possibly find better prediction equations of hydraulic parameters that are based not only on the lengths of pores, but also on their widths.

THEORY

We start with the equations for the concentrations of positive and negative univalent ions c_p and c_n (in mol m^{-3}), and the correspond-

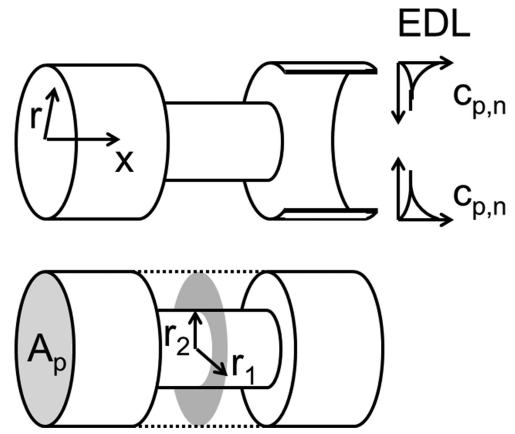


Figure 1. Schematic sketch of our model consisting of a sequence of wide and narrow pores. Top panel: definition of the coordinate system, and schematic behaviour of anomalous cation and anion concentration within the EDL. Bottom panel: definition of the pore section A_p to normalize the current, and definition of the radii r_1 and r_2 of wide and narrow pore, respectively.

ing current densities j_p and j_n (in A m^{-2}), in their 1-D form, as derived by Marshall & Madden (1959):

$$j_p = -F D_p \frac{\partial c_p}{\partial x} + F \mu_p c_p E, \quad (1)$$

$$j_n = -F D_n \frac{\partial c_n}{\partial x} - F \mu_n c_n E, \quad (2)$$

where F is Faraday's constant ($F = 96\,500 \text{ As mol}^{-1}$), D_p and D_n (in $\text{m}^2 \text{ s}^{-1}$) are diffusion coefficients of positive and negative ions, μ_p and μ_n (in $\text{m}^2 \text{ Vs}^{-1}$) denote the corresponding mobilities, respectively, and E is the electric field (in V m^{-1}). The x -direction is the direction along the pore (Fig. 1). The first term on the right-hand side describes the concentration diffusion, i.e. the tendency of ions to migrate from high to low concentration in order to maintain an equilibrium state. The second term on the right describes the transport of ions driven by the electric field. Note the difference in sign in the anion and cation equations: Both ion species tend to diffuse in the same direction (in opposite direction of the concentration gradients), but, due to the different signs of the ionic charge, in different directions when an electric field is applied.

Eqs (1) and (2) are coupled through Poisson's equation in the following form:

$$\frac{\partial E}{\partial x} = \frac{F}{\epsilon_0 \epsilon_r} (c_p - c_n), \quad (3)$$

where $\epsilon_0 = 8.85 \times 10^{-12} \text{ As Vm}^{-1}$ is the vacuum electrical permittivity and ϵ_r the relative permittivity. We used a typical value for water ($\epsilon_r = 80$) throughout this work.

In the model suggested by Marshall & Madden (1959), the induced polarization (IP) effect is generated by a variation of the mobilities along the pore. They assumed a cation selective zone where the current is mainly carried by the cations. The solution was then expressed in terms of cation and anion transference numbers t_p and t_n , representing the contribution of cations and anions to the total current, and defined as:

$$t_p = \frac{\mu_p}{\mu_p + \mu_n}, \quad (4)$$

$$t_n = \frac{\mu_n}{\mu_p + \mu_n}. \quad (5)$$

This definition results from a more general expression of transference numbers normally used in electrochemistry, which also depends on ion concentration and ion valence (e.g. Butt *et al.* 2003), by assuming univalent ions and the same concentration for both ion types. At the end of such a cation selective zone (in the direction of cation flow), cations will accumulate, and at the opposite side, a deficit will arise. Anions and cations are coupled through eq. (3). Even small concentration differences between the two ion species cause large electric field gradients, which will equilibrate anion and cation concentration. Essentially, $c_p = c_n$ can be assumed for low frequencies, which was also used by Marshall & Madden (1959) in their analytical solution and illustrated by Volkmann & Klitzsch (2010) based on numerical calculations. The ion selective zones cause an electrically neutral concentration gradient. The concentration gradients drive diffusion currents superposing the currents driven by the electric field, the net effect of which is an increase of resistance. All these processes are associated with a characteristic timescale. At high frequencies, the concentration gradients will have no time to build-up, at low frequencies they are fully expressed. The resistance of the pore will be frequency-dependent, constituting the IP-effect.

Marshall & Madden (1959) obtained equations of motion for the two ion species by taking the derivative with respect to the direction along the pore and applying the continuity equations:

$$F \frac{\partial c_p}{\partial t} + \frac{\partial j_p}{\partial x} = 0, \quad (6a)$$

$$F \frac{\partial c_n}{\partial t} + \frac{\partial j_n}{\partial x} = 0. \quad (6b)$$

They applied appropriate boundary conditions and linearized the equations system (eqs 1–3 and 6a,b) to obtain an analytical solution for the frequency-dependent impedance, hereafter called Marshall-Madden impedance, and discussed further below.

Extension to include pore radii

Marshall & Madden (1959) did not justify in detail why the transference numbers should vary along a pore, but it is natural to assume that the ion selectivity is related to the EDL. The (usually negative) charge at the surface of a mineral grain causes a fixed layer of (positive) ions attached to the mineral surface, called the SL (e.g. Schwarz 1962; Leroy *et al.* 2008). In the diffuse layer outside the SL, both ion species can move freely. Here, a cation surplus and an anion deficit develop which decrease with the distance to the pore surface (Fig. 1). The characteristic spatial scale is given by the Debye length λ_D (in m), which defines the distance from the surface of the SL, at which the charge deficit has decayed to the inverse of Euler's number of its original value:

$$\lambda_D = \sqrt{\frac{\epsilon_r \epsilon_0 k_B T}{2 c_0 e F}}. \quad (7)$$

Here, k_B is Boltzmann's constant (1.38×10^{-23} J K⁻¹), T is the temperature (in K), e is the electron charge (1.6×10^{-19} As) and c_0 (in mol m⁻³) is the equilibrium concentration in the free electrolyte.

In narrow pores, with a radius in the range of the Debye length, the cation surplus will be significant, causing the difference in transference numbers as postulated by Marshall & Madden (1959). Therefore, it seems self-evident to formulate a model where the cation selectivity is generated by a variation of pore radii instead of mobilities. Buchheim & Irmer (1979) suggested a similar approach

to account for the EDL by including concentrations averaged over the pore radii into the definition of transference numbers, without, however, explicitly formulating the theory.

In order to replace the parametrization in terms of varying mobility by one in terms of pore radii, we describe the pore space as a sequence of cylindrical units with different radii (Fig. 1). We define an average current density $\bar{j}_p$ by integrating over the cross-section of a pore A_p . From the cation eq. (1) we obtain

$$\bar{j}_p = \frac{1}{A_p} \int_{A_p} \left(-F D_p \frac{\partial c_p}{\partial x} + F \mu_p c_p E \right) dA \quad (8)$$

and analogously for the anions. Note that $\bar{j}_p$ is normalized to the section A_p of the larger of the two pores, to account for the fact that the total current will be limited by the pore section. In the smaller pore, the integrand is zero in the space between inner and outer cylinder.

The ion concentration depends on both x and r . We now assume that c_p and c_n can be factorized as:

$$c_{p,n}(x, r) = c'_{p,n}(x) \times b_{p,n}(r), \quad (9)$$

where the functions $b_{p,n}(r)$ (dimensionless) describe the variation of $c_{p,n}$ with distance from the pore surface, which are known from theories of the electric double layer used explicitly further below. The functions $c'_{p,n}(x)$ (in mol m⁻³) describe the variation of the concentration along the pore. Eq. (9) means that the variation along the pore and perpendicular to it do not depend on each other. This is a necessary step in order to be able to use the analytical solution with explicit parametrization of pore radii. To support the assumption, we argue that in a 2-D cylindrical pore the concentration is known to vary with distance from the pore surface, and the variation can be described with a functional form. The concentration variation in radial direction is significantly stronger (orders of magnitude) than that in the x -direction (just a small fraction of one, e.g. Blaschek & Hördt 2009). The influence, for example, of a $c'_p(x)$ variation on the $b_p(r)$ shape will be a second-order effect, probably most serious in the vicinity of the contact zone between two pores. The alternative, a full solution of the 3-D equations for particular pore systems, is far more complicated and only possible numerically. In the results section, we will discuss a comparison of our results with such a numerical solution.

In the following, we use the Nernst–Einstein relationship between diffusion constant and mobility:

$$D_p = \frac{\mu_p k_B T}{e}. \quad (10)$$

We write out the pore radii r_1 and r_2 explicitly, via

$$A_p = \pi r_1^2. \quad (11)$$

Making use of the radial symmetry and inserting eq. (9) into eq. (8), we obtain

$$\bar{j}_p = \frac{1}{\pi r_1^2} \int_0^{r_1} \left(-F \frac{\mu_p k_B T}{e} \frac{\partial (c'_p(x) b_p(r))}{\partial x} + F \mu_p c'_p(x) b_p(r) E \right) \times 2\pi r dr \quad (12)$$

in the larger pore, and analogously in the smaller pore, where the integration limit is r_2 , and the normalization is still carried out with r_1 . Rearrangement yields

$$\bar{j}_p = \left(-F \frac{\mu_p k_B T}{e} \frac{\partial c'_p(x)}{\partial x} + F \mu_p c'_p(x) E \right) \frac{1}{\pi r_1^2} \int_0^{r_1} b_p(r) 2\pi r dr, \quad (13)$$

which may also be written as

$$\bar{j}_p = -F \frac{\mu_p \bar{b}_p k_B T}{e} \frac{\partial c'_p(x)}{\partial x} + F \mu_p \bar{b}_p c'_p(x) E, \quad (14)$$

where $\bar{b}_p$ was defined as

$$\bar{b}_p = \frac{1}{\pi r_1^2} \int_0^{r_1} b_p(r) 2\pi r dr. \quad (15)$$

The derivation of the corresponding anion equation is completely analogous.

In eq. (14), the current density is expressed as function of the concentration gradient and the product of concentration and electric field in the same form as in eq. (1). Therefore, we may use the same solution for the impedance, but with $\bar{b}_p$ (and $\bar{b}_n$) as additional parameters. The dimensionless numbers $\bar{b}_p$ and $\bar{b}_n$ represent average ion concentrations, normalized to the concentration c_0 of the free electrolyte. They include the geometrical information, and thus the cation-selective properties of the narrow pores. It is no longer necessary to postulate the cation-selective properties and parametrize them in terms of mobility, since they come out naturally as a result of the difference in pore radius in relationship with the EDL properties.

From eq. (14), we see that the transference numbers can easily be extended to include geometry, because the parameters $\bar{b}_p$ (and $\bar{b}_n$) and the mobility occur as a product in both terms. Therefore, we define averaged transference numbers as:

$$\bar{t}_p = \frac{\mu_p \bar{b}_p}{\mu_p \bar{b}_p + \mu_n \bar{b}_n}, \quad (16)$$

$$\bar{t}_n = \frac{\mu_n \bar{b}_n}{\mu_p \bar{b}_p + \mu_n \bar{b}_n}. \quad (17)$$

These averaged transference numbers are now consistent with the conventional definition, because they include ion concentrations, the calculation of which will be explained in the next section.

Calculation of ion concentration

The calculation of the parameters $\bar{b}_{p,n}$ is equivalent to the calculation of ion concentration with distance from the mineral surface, which is based on the electrical potential. For a cylindrical pore with radius r_1 , the potential φ_{cyl} (in V) is given by (e.g. Hunter 1981)

$$\varphi_{cyl}(r) = \zeta \frac{J_0(j\kappa r)}{J_0(j\kappa r_1)}, \quad (18)$$

where $j^2 = -1$, J_0 is the zero-order Bessel function, and κ , the Debye parameter, is defined as the inverse of the Debye length as given in eq. (7). The Zeta potential ζ is the electrical potential at the surface of shear, an 'imaginary surface which is considered to lie close to the solid surface and within which the fluid is stationary' (Hunter 1981). In the model by Leroy & Revil (2004), it corresponds to the interface between the SL and the diffusive layer. Eq. (18) describes the potential from inside the pore to the surface of shear. The zone between the surface of shear and the surface of the mineral grain (the SL) will be considered in form of a correction factor.

The ion concentrations, normalized by the concentration in the free electrolyte c_0 , are then calculated from (e.g. Butt *et al.* 2003).

$$b_{n,p}(r) = e^{\pm \frac{e\varphi_{cyl}(r)}{k_B T}}, \quad (19)$$

Note that there is an asymmetry in b_p and b_n caused by the different sign of the exponent in eq. (19), which reflects, depending on the Zeta potential, the increase of the concentration of one ion species and the decrease of the concentration of the other ion species within the diffuse layer.

The approach defined by eqs (18) and (19) describes the ion concentration in the diffuse layer. The SL also contributes to the cation concentration in the EDL, but is not included in the above formalism because the cations are directly attached to the mineral surface and do not follow the differential equation describing the distribution of free ions. The contribution of the ions in the SL to the overall conductivity is a subject of current debate. For example Gimmi & Kosakowski (2011) stressed the unresolved contribution of SL cations to surface mobility values when presenting their model for surface diffusion of sorbed cations. In the absence of a generally accepted theoretical model to include the SL contribution, we follow the suggestion of Leroy & Revil (2004) and Leroy *et al.* (2008), who introduce the surplus charge per unit area σ (in As m⁻²) parallel to the pore surface. These authors define σ^d and σ^s as the charge densities in the diffuse and in the SL, respectively, and the partition coefficient f_Q as the contribution of the SL to the total surplus charge:

$$f_Q = \frac{\sigma^s}{\sigma^s + \sigma^d}. \quad (20)$$

The total surplus charge density, i.e. the sum of σ^d and σ^s , must be equal to the surface charge density of the mineral, and depends on electrochemical processes at the mineral surface, the pH and ion species of the electrolyte. More detailed treatments including tables with typical values for partition coefficients can be found in Leroy *et al.* (2008) and Leroy & Revil (2009).

Using the partition coefficient as defined in eq. (20), it can be shown (see Appendix A) that the contribution of the SL may be considered in the form of a correction to the average cation concentration

$$\tilde{b}_p = \frac{\bar{b}_p - f_Q}{1 - f_Q}. \quad (21)$$

Since $\bar{b}_p$ is larger than 1, $\tilde{b}_p$ increases monotonously with f_Q . The average cation concentration is increased by the correction, because the cations in the SL, which were so far ignored, are now added to the overall concentration. For small f_Q , corresponding to a small SL, the correction becomes negligible. The corrected form (eq. 21) will be used in the averaged transference numbers (eqs (16) and (17)), which completes our model. The full impedance equation according to Marshall & Madden (1959) with our extension and notation is given in Appendix B. The theory has been worked out here for two univalent ions, but could easily be extended to the case of several different ion types.

The pore radii will have the most notable effect on the parameters $\bar{b}_{p,n}$, because they enter with the power of two into eq. (15). However, all other parameters determining the structure of the EDL, such as temperature, equilibrium concentration of the free electrolyte and the potential at the mineral surface, enter as well through the Debye-length (eqs 7 and 18).

COMPARISON WITH NUMERICAL SIMULATION

Since our concept involves some assumptions, we evaluate its validity by comparison with numerical simulation results. The work

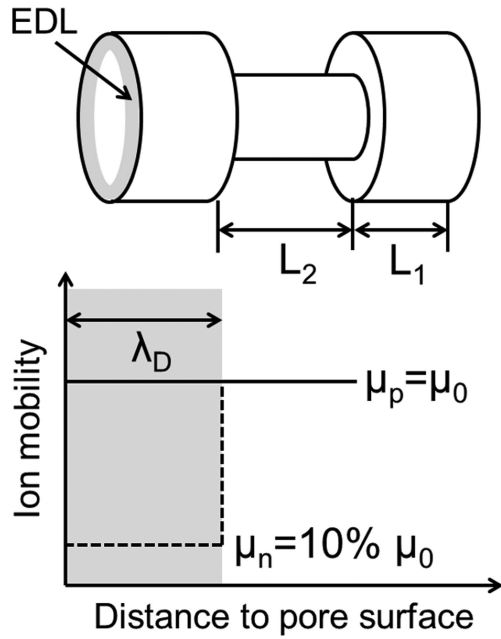


Figure 2. Sketch of the model defined by Volkmann & Klitzsch (2010) to simulate membrane polarization. Top panel: cylindrical model with pore lengths $L_1 = 5 \mu\text{m}$, $L_2 = 0.05 \mu\text{m}$. Bottom panel: definition of the outer shell with anion mobility within the EDL reduced to 10 percent compared to the free electrolyte. The pore radii are $r_1 = 0.5 \mu\text{m}$, $r_2 = 0.05 \mu\text{m}$, respectively, and the mobility in the free electrolyte is $5 \times 10^{-8} \text{ m}^2 \text{ Vs}^{-1}$.

most suitable for this purpose seems to be Volkmann & Klitzsch (2010), who modelled membrane polarization by solving the 3-D equivalent of the underlying equation system (eqs 1–3) with a commercial finite element package. Fig. 2 shows a sketch of their model we chose for comparison. It consists of two cylindrical pore types with different radii. Each pore consists of two zones, where the outer shell describes the EDL, the inner cylinder the free electrolyte. Due to restrictions in the spatial resolution, it was not possible to account for the EDL by assuming a fixed potential at the mineral surface. Instead, the EDL was simulated by assuming a layer of finite thickness, equal to the Debye length, where the anion and cation mobilities differ.

In order to find an equivalent model with our concept that should yield comparable results, we replace the change of mobilities by a change of ion concentrations (Fig. 3). Close to the pore surface, there is an anion deficit, whereas equilibrium is assumed in the inner cylinder. The parameters $\bar{b}_{p,n}$ defined above are calculated such that

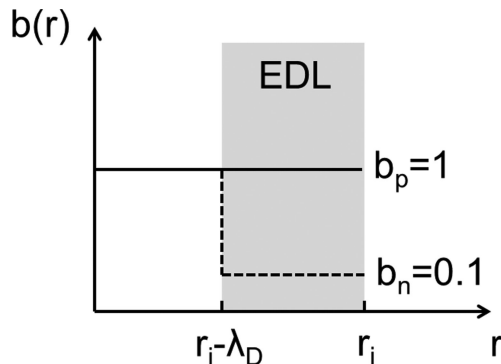


Figure 3. Sketch of the model to simulate a layered EDL, equivalent to the one described in Fig. 2. The EDL is simulated by a reduced anion concentration compared to the concentration of the free electrolyte c_0 .

the transport numbers in the two pores are the same as those in the model by Volkmann and Klitzsch. The evaluation of eq. (15) for the geometry shown in Fig. 3 yields for $\bar{b}_{p,n}$

$$\bar{b}_{pi} = 1, \quad (22)$$

$$\bar{b}_{ni} = \frac{1}{r_i^2} (r_i^2 - 0.9 (2\lambda_D r_i - \lambda_D^2)), \quad (23)$$

where the index i denotes the two pore types of different size, with $i = 1$ being the wider pore. Note that the normalization with the cross-section of the larger pore is considered in eq. (23) by using r_1 in the denominator. The corresponding averaged transference numbers are then simply obtained from eqs (16) and (17); the correction for the SL is obsolete in this case.

Several model parameters may be varied, and different parameters extracted from the resulting spectra may be used to compare the numerical results with our analytical approximation. Here, we chose the phase shifts with the maximum absolute value as a measure of the polarization effect, displayed as function of the pore radii, which is actually the new feature that can be modelled with our approach. The numerical solution and our approach agree qualitatively in the sense that the polarization first increases and then decreases again with increasing pore radius, for both ion concentrations (Fig. 4, top panel). The minimum phase values also agree

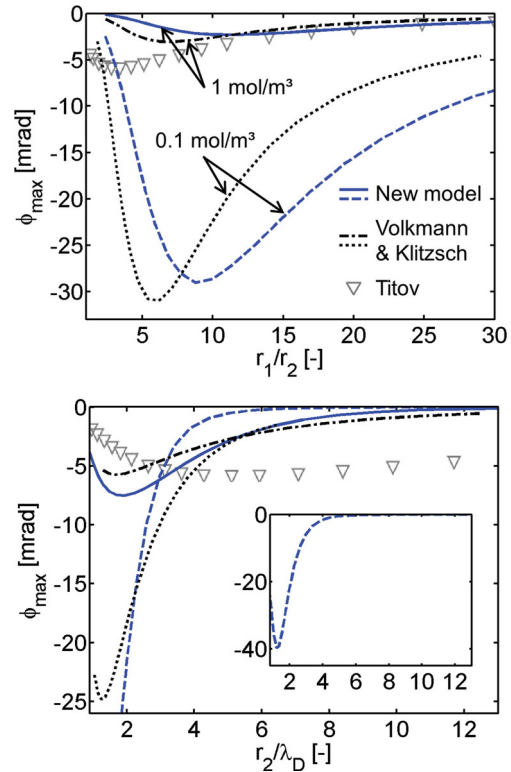


Figure 4. Phase shift with maximum absolute value of the impedance calculated for the model shown in Fig. 2 with the numerical approach (black dash-dot and dotted line) by Volkmann & Klitzsch (2010), and by our analytical approach for the model defined in Fig. 3 (blue solid and dashed lines). Top panel: maximum phase shift versus ratio of pore radii (r_1 is varied), for two different concentrations of the free electrolyte. Bottom panel: maximum phase shift versus radius of the smaller pore, normalized by the EDL thickness. The inset shows the dashed line with a different scaling. The triangles denote an attempt by Volkmann & Klitzsch (2010) to compare their results with a model by Titov *et al.* (2002).

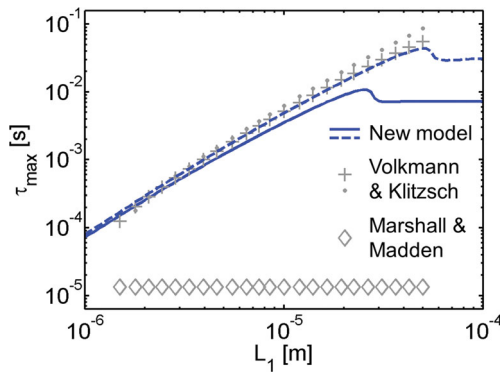


Figure 5. Relaxation time (inverse of angular frequency of maximum phase shift), versus length of the wider pore L_1 . Radii and mobilities are as given in Fig. 2. Dots and crosses: Numerical results from Volkman & Klitzsch (2010) for $c_0 = 0.1 \text{ mol m}^{-3}$ and $c_0 = 1 \text{ mol m}^{-3}$, respectively. Diamonds: Marshall–Madden impedance without expansion, as calculated by Volkman & Klitzsch (2010). Lines: expanded Marshall–Madden impedance for $c_0 = 0.1 \text{ mol m}^{-3}$ (dashed) and 1 mol m^{-3} (solid).

very well (e.g. approx. -30 mrad for 0.1 mol m^{-3}), whereas there is some difference in the radius at which the minimum is observed. On the other hand, the agreement between the results of Volkman & Klitzsch (2010) and the results obtained with the so-called ‘short narrow pore (SNP)’ model by Titov *et al.* (2002) is poor, indicating that assumptions of the SNP-model are poorly fulfilled.

The qualitative agreement between the numerical solution and our model also holds as function of the smaller pore radius (Fig. 4, bottom panel). The polarization adopts a maximum if the smaller pore radius is in the range of the Debye length (10^{-8} m for the chosen parameters), and then quickly decreases. For the smaller ion concentration, there is a difference of approx. 50 per cent in the minimum phase shifts between the numerical and our solution. Again, the comparison with the SNP-model is considerably more difficult.

The relaxation times obtained from the numerical simulation also agree very well with those obtained from our expanded Marshall–Madden impedance (Fig. 5). The relaxation times are defined by

$$\tau_{\max} = \frac{1}{\omega_{\max}}, \quad (24)$$

where $\omega_{\max}$ is the angular frequency at which the maximum phase shift occurs. The length of the wide pore is varied, because this parameter yields the largest dynamics of $\tau_{\max}$ for this particular set of parameters. The comparison of the relaxation times by Volkman & Klitzsch (2010) with those obtained from the conventional Marshall–Madden impedance is apparently difficult because pore radii cannot be included. We also conclude from Fig. 5 that the relaxation times are only weakly dependent on salinity: the curves for the two concentrations differ only slightly, consistent with experimental observations (e.g. Weller *et al.* 2011).

The causes of the differences between our approach and the numerical solution are not exactly known. A plausible candidate may be our basic assumption that concentrations in radial and axial directions are independent (eq. 9), which basically ignores 2-D and 3-D effects. Secondly, the numerical solution will neither be perfect. We conclude from this study that our concept is feasible at least for studies of the qualitative behaviour of the impedances. It may be used for quantitative estimates as well; in particular the relaxation times seem to be consistent.

The model used for comparison with numerical simulations (Fig. 3) may be oversimplified, not only because the EDL is de-

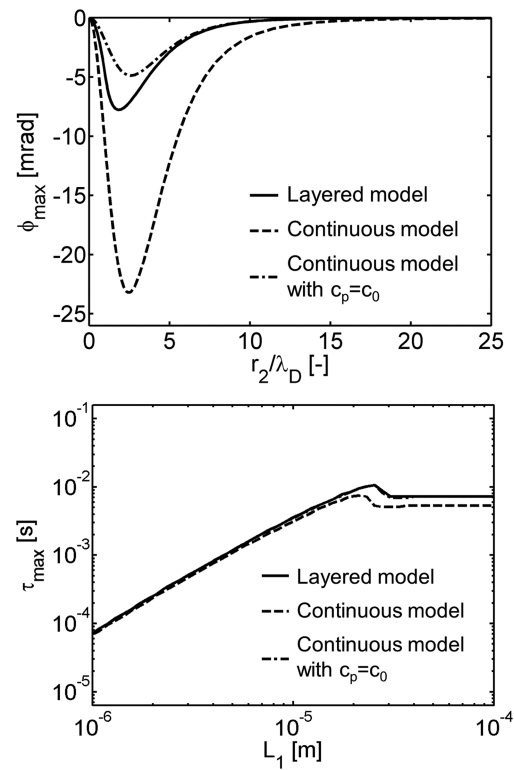


Figure 6. Maximum phase shift and relaxation time calculated with our new model. In both panels, the solid line was calculated to approximate the layered model shown in Fig. 3 and is identical to the solid line in Figs 4 (bottom panel) and 5, respectively, for $c_0 = 1 \text{ mol m}^{-3}$. The dashed and dash-dotted lines were calculated for more realistic continuous double layer models, further explained in the main section. Top panel: phase shift with maximum absolute value versus normalized radius of the smaller pore. Bottom panel: relaxation time versus length of the wider pore.

scribed as a discrete layer, but also because only the anion transference number is varied. We carry out the transition from the layered model to a model with a realistic continuous EDL as described by eqs (18) and (19) in two steps (Fig. 6). First, instead of forcing the EDL thickness and the concentrations to certain values, as in Fig. 3, we use the Zeta potential as a free parameter. The Zeta potential is determined such that the anion concentration averaged over the pore section is the same as that defined by Fig. 3 with the corresponding parameters for c_0 (1 mol m^{-3}) and μ ($5 \times 10^{-8} \text{ m}^2 \text{ Vs}^{-1}$). Assuming that the radius of the pore is large compared to the Debye length, we can determine a single value for the Zeta potential, independent of the pore radius ($\zeta = -0.0293 \text{ V}$ in this case). If the assumption of a large pore radius is violated, the concentration will not relax to c_0 in the middle of the pore and the equivalent Zeta potential will depend on the pore radius. The cation concentration is still kept constant at $c_p = c_0$ in this first step. The difference between this model and the layered model is small (dash dotted lines in Fig. 6), except for small pore radii (Fig. 6, top panel), which is due to the violation of the assumption of a large pore radius.

In a second step, we use the Zeta potential calculated above, but now allow the cation concentration to vary as well. The variation of cation concentration has a large impact on the maximum phase shifts (Fig. 6, top panel). The influence on the relaxation times (bottom panel) is smaller, and limited to relatively long narrow pores. We conclude that a layered EDL model may constitute a useful approximation that does not produce dramatic errors compared to a

continuous EDL model. For quantitative estimates and small pore radii, however, the continuous model should be used.

PARAMETER STUDIES

The inclusion of pore radii as new parameters in the analytical impedance formulation renders systematic parameter studies possible, to investigate under which conditions membrane polarization might make a significant contribution to the overall polarization of natural media. With numerical simulations, only a limited parameter regime can be exploited for practical or technical reasons, such as computing time and discretization issues (e.g. Blaschek & Hördt 2009; Volkmann & Klitzsch 2010). From previous studies it is evident that the parameter space of the Marshall–Madden impedance has different regimes in which different relationships between, for example, relaxation times and geometrical parameters exist. Thus, important relationships might be overlooked when investigating limited regions.

Here, we have further increased the number of parameters by the properties of the EDL and the radii of the two pores, and it is not trivial to derive general relationships or to fully exploit the parameter space. Therefore, we only discuss two examples here to demonstrate the new possibilities of our expansion, and leave a systematic analysis for future work.

First, we display the dependence of the IP parameters (maximum phase shift and relaxation time) on the radii of the two pores (Fig. 7). This is one of the new possibilities compared to previous analytical studies, where the geometrical parameters are usually the lengths of pores, and not their widths. All other parameters are fixed and given in the figure caption. This time, we apply the model with the continuous EDL (eq. 19), and thus there is no exact correspondence with any model by Volkmann & Klitzsch (2010). However, the parameters were chosen such that the transference numbers are comparable, i.e. the parameters for their figs 15 and 16 would each form a horizontal line in our figure, crossing in the centre.

The phase shift adopts a maximum in a region where the ratio between the two pore radii is approximately 10:1. On this diagonal, there is a maximum if the radius of the smaller pore is approx. equal to the Debye length (10^{-8} m for the chosen parameters), which is a measure of the thickness of the EDL. This means that maximum phase shifts should be observed for extremely small pore radii. The phase values are relatively large, with a maximum in the order of 100 mrad, and even for larger pore radii considerable phase shifts in the mrad range are obtained.

The relaxation time (bottom panel) also depends on the pore radii. This fact is important in itself, because in the original Marshall–Madden impedance, there is only a weak dependence of relaxation time on the mobility contrast (e.g. Blaschek & Hördt 2009), which might imply that the dependence on pore radii is also weak. In fact, the largest variation of relaxation time occurs where the radii of the two pores are approximately equal, which is the range where the two pores change their role as narrow and wide pore. In this region, the IP-effect itself (measured by the maximum phase shift) is weak (Fig. 7, top panel), and thus the dependence of relaxation times might be considered somewhat artificial. However, the relationships are apparently not simple, as also indicated by the broken symmetry in the lower left corner of Fig. 7 (bottom panel) which is caused by the asymmetry in pore lengths.

Our second parameter study shows that the ratio of pore radii that provides the maximum phase shift depends on the length ratio (Fig. 8, top panel). The line of maximum phase shift indicated in

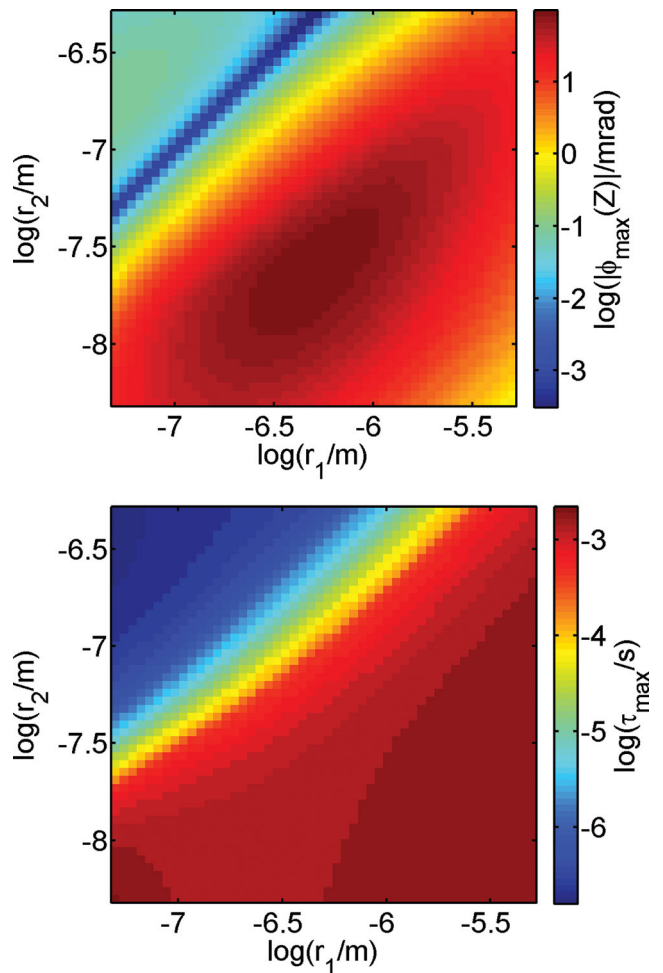


Figure 7. Top panel: maximum phase shift, colour coded with logarithmic colour scale as function of the two pore radii, for the model sketched in Fig. 1. Bottom panel: relaxation time, calculated as the inverse of the angular frequency at which the maximum phase shift occurs. The pore lengths are $L_1 = 5 \mu\text{m}$, $L_2 = 0.05 \mu\text{m}$, the mobility and the ion concentration in the free electrolyte are $5 \times 10^{-8} \text{ m}^2 \text{ Vs}^{-1}$ and 1 mol m^{-3} , respectively. The partition coefficient is $f_Q = 0.6$ and the Zeta potential is -75 mV .

the figure is defined by

$$\frac{r_1}{r_2} = \sqrt{\frac{L_1}{L_2}}. \quad (25)$$

Obviously, the 10:1 ratio of the pore radii found in Fig. 7 is a specific case that was derived for the length ratio 100:1. The relationship given by eq. (25) may not be generally valid, because the radius and length of the second pore are fixed, and the figure may depend on them. A full generalization is difficult, because the Debye length introduces an absolute length scale.

Most of the phase shift values in Fig. 8 are out of the currently measurable range of approx. 0.1 mrad, and only the dark red section in the lower left corner would be considered relevant from a practical point of view. We also observe that point symmetry is lacking in the figure. When crossing the horizontal line $r_1 = r_2$, the two pores apparently change their role, and one might expect point symmetry with respect to the point (0,0) in this double logarithmic scale. The symmetry is broken because we had to choose a fixed value for L_2 , and the geometrical parameters are all associated with an absolute value (the Debye length).

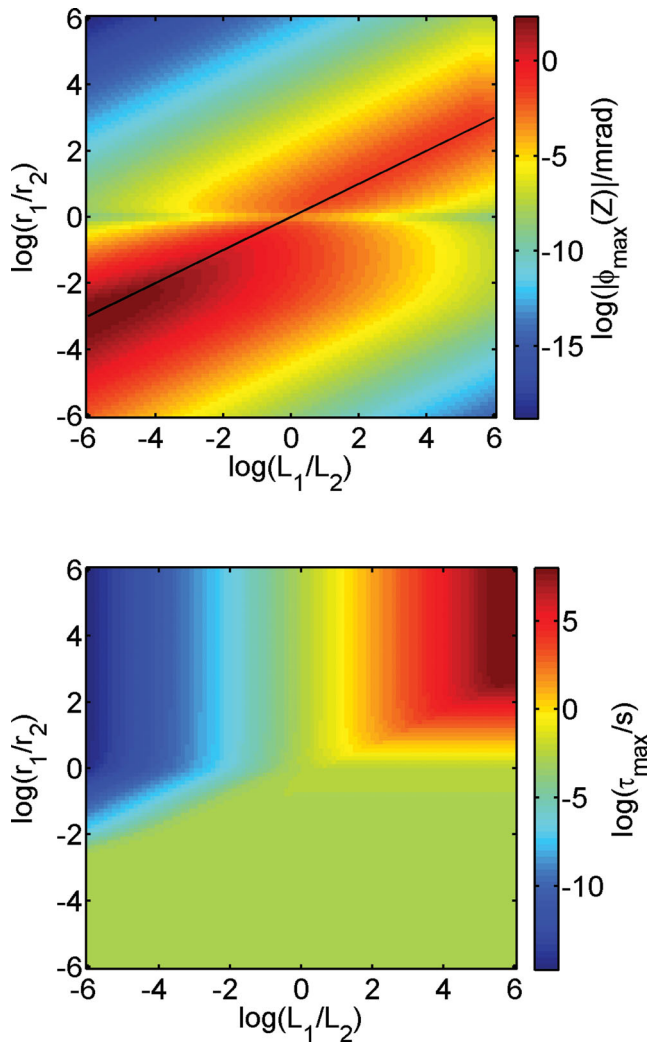


Figure 8. Top panel: maximum phase shift, colour coded with logarithmic colour scale as function of pore radii ratio and the pore length ratio, for the model sketched in Fig. 1. The line indicates the maximum phase shift. Bottom panel: relaxation time, calculated as the inverse of the angular frequency at which the maximum phase shift occurs. The pore radius r_2 is fixed to $r_2 = 5 \times 10^{-6}$ m, whereas r_1 is varied, and the pore length L_2 is fixed to $L_2 = 5 \times 10^{-6}$ m. The mobility and the ion concentration in the free electrolyte are 5×10^{-8} ($\text{m}^2 \text{Vs}^{-1}$) and 1 mol m^{-3} , respectively. The partition coefficient is $f_Q = 0$ and the Zeta potential is -75 mV .

The relaxation times exhibit different regimes characterized by different dependencies on the geometrical parameters (Fig. 8, bottom panel). For narrow pores (small r_1), relaxation times are independent of pore radii and pore length L_1 . The relaxation times depend on the length of the wide pore (L_2), which does not cause any observable variation because L_2 was fixed. For large radii r_1 (top section of Fig. 8, bottom panel), relaxation times depend on L_1 because the pores changed role, and pore 1 is the wide pore. The relaxation times depend on pore radii preferably in the transition zone around the horizontal line $r_1 = r_2$, consistent with the finding from the previous discussion of Fig. 7. The role of the transition zone can be understood from the behaviour of the Marshall–Madden impedance. It can be shown (Bücker 2011) that each pore is associated with its own relaxation time, the shorter of which dominates the impedance behaviour. In the transition zone, where the pores change role, the dominating relaxation time will also change. The phase shifts in the transition region are small, and the range where

this dependence is relevant in practice might also be small. We can conclude that while the relaxation times are independent of pore widths in certain parameter regimes, the independence does not hold as a general rule.

CONCLUSIONS

We have suggested an extension of the 1-D Marshall–Madden impedance to include the properties of the EDL and the pore radii. The inevitable, but well justified assumption is that the ion concentrations in the directions parallel and perpendicular to the axis of the cylindrical pore system are independent of each other. This ignores 2-D effects, which will be significant in particular in the vicinity of contact zones between the pores. Our comparison with numerical simulation results indicates that this does not cause dramatic errors in the impedance.

Our approach will be useful to verify the feasibility of numerical simulations at least qualitatively. Since the required spatial scales may span several orders of magnitude, the discretization of the equation system can be an issue, and the possibility to compare the results for simple models with an analytical solution may be helpful.

The extension opens up a wide range of new simulation possibilities. A much wider parameter range can be explored, which should compensate for the minor drawback that 2-D effects have to be ignored. The parameter space has become quite large: in addition to the two pore lengths and two pore radii, the equilibrium concentration and the ion mobilities, the parameters characterizing the EDL, i.e. Zeta potential (which depends on further parameters, such as pH), temperature, and partition coefficient, can also be considered in the model. In spite of the large number of parameters that can be varied, it should be possible through systematic studies, analytical derivations and scaling laws to increase the general understanding of membrane polarization and to derive general conclusions for the behaviour of the impedance.

The parameter studies indicate that the relaxation time derived from the impedance spectra may depend on the pore radii. This might be important for the prediction of hydraulic conductivity. The importance of using pore throat size in hydraulic conductivity prediction was also recognized by Revil *et al.* (2012) who suggested a new relationship that includes the dynamic pore radius rather than grain diameter, without a related mechanistic model, however. Kruschwitz *et al.* (2010) hypothesized that pore throat size controls different regimes of relaxation times. Our second preliminary conclusion is that the maximum polarization occurs when the narrower pore has the size of the Debye length and the ratio of the pore widths equals the square root of the pore length ratio.

Finally, our theory might constitute a step towards merging grain-based and pore space-based theories. The explicit consideration of the EDL properties in a membrane polarization model provides a connection between the two types of theories that might be exploited in future work.

ACKNOWLEDGEMENTS

We would like to thank Norbert Klitzsch and Jan Volkmann for fruitful discussion of this work, and for providing their data displayed in Figs 4 and 5. Christian Fulda gave some useful hints that improved the original version of the manuscript. We thank Damien Jougnot and an anonymous reviewer for their deep and helpful comments. The work was carried out within the gebo research association, sponsored by Baker Hughes and the Ministry of Science and Culture of Lower Saxony, Germany.

REFERENCES

- Blaschek, R. & Hördt, A., 2009. Numerical modelling of the IP-effect at the pore scale, *Near Surf. Geophys.*, **7**, 579–588.
- Buchheim, W. & Irmer, G., 1979. Zur Theorie der induzierten galvanischen Polarisation in Festkörpern mit elektrolytischer Porenfüllung, *Gerlands Beitr. Geophysik*, **88**, 53–73.
- Bucker, M., 2011. Die Membranpolarisation als Ursache des IP-Effekts in porösen Medien, *MSc thesis*, TU Braunschweig.
- Butt, H.-J., Graf, K. & Kappel, M., 2003, *Physics and Chemistry of Interfaces*, Wiley-VCH, Weinheim.
- Gimmi, T. & Kosakowski, G., 2011. How mobile are sorbed cations in clays and clay rocks? *Environ. Sci. Technol.*, **45**, 1443–1449.
- Hördt, A., Blaschek, R., Binot, F., Druventak, A., Kemna, A., Kreye, P. & Zisser, N., 2009. Case histories of hydraulic conductivity estimation with induced polarization at the field scale, *Near Surface Geophys.*, **7**, 529–545.
- Hördt, A. & Milde, S., 2012. Studies on the origin of induced polarisation with gel-filled sandstone samples, *Near Surface Geophys.*, **6**, 469–478.
- Hunter, R.J., 1981. *Zeta Potential in Colloid Science: Principles and Applications*, Academic Press, London.
- Jougnot, D., Ghorbani, A., Revil, A., Leroy, P. & Cosenza, P., 2010. Spectral induced polarization of partially saturated clay-rocks: a mechanistic approach, *Geophys. J. Int.*, **180**, 210–224.
- Juang, C.H. & Holtz, R.D., 1986. A probabilistic permeability model and the pore size density function, *Int. J. Numer. Anal. Methods Geomech.*, **10**, 543–553.
- Kemna, A., Binley, A. & Slater, L., 2004. Crosshole IP imaging for engineering and environmental applications, *Geophysics*, **69**, 97–107.
- Kemna, A., Münch, H.M., Titov, K., Zimmermann, E. & Vereecken, H., 2005. Relation of SIP relaxation time of sands to salinity, grain size and hydraulic conductivity, in *Proceedings of the 11th Eur. Meeting Environ. Eng. Geophys., Exp. Abst.*, European Association of Geoscientists & Engineers (EAGE), Palermo, Italy.
- Koch, K., Revil, A. & Holliger, K., 2012. Relating the permeability of quartz sands to their grain size and spectral induced polarization characteristics, *Geophys. J. Int.*, **190**, 230–242.
- Kruschwitz, S., Binley, A., Lesmes, D. & Elshenawy, A., 2010. Textural controls on low-frequency electrical spectra of porous media, *Geophysics*, **75**, WA113–WA123.
- Leroy, P. & Revil, A., 2004. A triple-layer model of the surface electrochemical properties of clay minerals, *J. Colloid Interface Sci.*, **270**, 371–380.
- Leroy, P. & Revil, A., 2009. A mechanistic model for the spectral induced polarization of clay materials, *J. geophys. Res.*, **114**, B10202, doi:10.1029/2008JB006114.
- Leroy, P., Revil, A., Kemna, A., Cosenza, A. & Ghorbani, A., 2008. Complex conductivity of water-saturated packs of glass beads, *J. Colloid Interface Sci.*, **321**, 103–117.
- Marshall, D.J. & Madden, T.R., 1959. Induced polarization: a study of its causes, *Geophysics*, **24**, 780–816.
- Revil, A. & Florsch, N., 2010. Determination of permeability from spectral induced polarization in granular media, *Geophys. J. Int.*, **181**, 1480–1498.
- Revil, A., Koch, K. & Holliger, K., 2012. Is it the grain size or the characteristic pore size that controls the induced polarization relaxation time of clean sands and sandstones? *Water Res. Res.*, **48**, W005602, doi:10.1029/211WR011561.
- Schwarz, G., 1962. A theory of the low-frequency dielectric dispersion of colloidal particles in electrolyte solution, *J. Phys. Chem.*, **66**, 2636–2642.
- Scott, J.B., 2006. The origin of the observed low-frequency electrical polarization in sandstones, *Geophysics*, **71**, 235–238.
- Sidhorova, M. & Fridrikhsberg, D., 1978. Study of induced polarization of systems simulating soils during freezing, in *Proceedings of the Permafrost: The USSR Contribution to the Second International Conference*, National Academy of Sciences, pp. 347–349.
- Titov, K., Komarov, V., Tarasov, V. & Levitski, A., 2002. Theoretical and experimental study of time domain-induced polarization in water-saturated sands, *J. Appl. Geophys.*, **50**, 417–433.
- Titov, K., Tarasov, A., Ilyin, Y., Seleznev, N. & Boyd, A., 2010. Relationships between induced polarization relaxation time and hydraulic properties of sandstone, *Geophys. J. Int.*, **180**, 1095–1106.
- Volkman, J. & Klitzsch, N., 2010. Frequency-dependent electric properties of microscale rock models for frequencies from one millihertz to ten kilohertz, *Vadoze Zone J.*, **9**, 858–870.
- Weller, A., Nordsiek, S. & Debschütz, W., 2010a. Estimating permeability of sandstone samples by nuclear magnetic resonance and spectral-induced polarization, *Geophysics*, **75**, E215–E226.
- Weller, A., Slater, L. & Nordsiek, S., 2010b. On the estimation of specific surface per unit volume from induced polarization: a robust empirical relation fits multiple data sets, *Geophysics*, **75**, WA105–WA112.
- Weller, A., Breede, K., Slater, L. & Nordsiek, S., 2011. Effect of changing water salinity on complex conductivity spectra of sandstones, *Geophysics*, **76**, F315–F327.

APPENDIX A: STERN LAYER CORRECTION

In order to derive the correction for the contribution to the Stern layer (SL), we first relate the surplus charge per unit area in the diffuse layer σ^d to the normalized average cation concentration $\tilde{b}_p$

$$2\pi r \sigma^d = F c_0 A_p (\tilde{b}_p - 1). \quad (A1)$$

The left-hand side is a surplus charge per unit length, where the length is in the direction of the pore. The right-hand side is the normalized surplus ion density ($\tilde{b}_p - 1$) multiplied with F and c_0 to convert it into a charge density, and with the pore section to convert it into a charge per unit length density. In the corrected form, the contribution of the SL is considered as well, and thus we can write the equation for the corrected normalized cation concentration $\tilde{b}_p$

$$2\pi r (\sigma^d + \sigma^s) = F c_0 A_p (\tilde{b}_p - 1). \quad (A2)$$

Division of eq. (A2) by (A1) yields:

$$\frac{\tilde{b}_p - 1}{\tilde{b}_p - 1} = \frac{\sigma^d + \sigma^s}{\sigma^d}. \quad (A3)$$

The final form (eq. (21) of the main section) is obtained after some algebra by solving for $\tilde{b}_p$ and using the definition of the partition coefficient (eq. 20).

APPENDIX B: EXTENDED MARSHALL-MADDEN IMPEDANCE

The impedance derived by Marshall & Madden (1959) for a 1-D pore system, extended for the 2-D system defined in Figs 1 and 2, writes

$$Z(\omega) = \frac{L_1}{\mu_{p1} \tilde{b}_{p1} c_0 F} \left[\tilde{t}_{p1} + \frac{\tilde{B}}{A} \tilde{t}_{p2} + \frac{(\tilde{S}_2 - \tilde{S}_1)^2}{\frac{\tilde{X}_1(\omega) \tilde{S}_1}{\tilde{t}_{p2}^2 \tilde{t}_{p1} \tanh(\tilde{X}_1(\omega))} + \frac{A}{B} \frac{\tilde{X}_2(\omega) \tilde{S}_2}{\tilde{t}_{p1}^2 \tilde{t}_{p2} \tanh(\tilde{X}_2(\omega))}} \right], \quad (B1)$$

where

$$\tilde{X}_i(\omega) = \left(\frac{j\omega}{2D_{pi} \tilde{t}_{ni} \tilde{b}_{pi}} \right)^{\frac{1}{2}} \frac{L_i}{2}. \quad (B2)$$

This is the same notation as that used in the original work, extended by the tilde to indicate values averaged over the pore section, and the

correction for the SL. Parameter A is the length ratio, and remains unchanged compared to the original definition:

$$A = \frac{L_1}{L_2}. \quad (\text{B3})$$

Parameter $\tilde{B}$ is the ratio of diffusivities, multiplied with the ratio of averaged, normalized and corrected concentrations

$$\tilde{B} = \frac{D_{p1}}{D_{p2}} \frac{\tilde{b}_{p1}}{\tilde{b}_{p2}}. \quad (\text{B4})$$

$\tilde{S}$ is the ratio of anion and cation transference numbers in each of the two zones

$$\tilde{S}_i = \frac{\tilde{t}_{ni}}{\tilde{t}_{pi}}. \quad (\text{B5})$$

The transference numbers are given by

$$\tilde{t}_{p,ni} = \frac{\mu_{p,ni} \tilde{b}_{p,ni}}{\mu_{pi} \tilde{b}_{pi} + \mu_{ni} \tilde{b}_{ni}} \quad (\text{B6})$$

with

$$\tilde{b}_{pi} = \frac{\bar{b}_{pi} - f_Q}{1 - f_Q}, \quad (\text{B7})$$

whereas

$$\tilde{b}_{ni} = \bar{b}_{ni} \quad (\text{B8})$$

because the anion concentration is unaffected by the SL.

In the examples shown in this work, we assume that diffusivities (and mobilities) are equal in the two zones and for the two ion species, i.e.

$$D_{pi} = D_{ni} = D_{0i} \quad (\text{B9})$$

$$D_{01} = D_{02} = D_0 \quad (\text{B10})$$

and analogously for the mobilities, which simplifies some of the equations. We leave them here in the general form to maintain the option to consider varying diffusivities.